

_Program Question_4

Implicit none

Real::x0,x1,x2,x,f_x0,f_x1,f_x2,net_error

integer::n

x0=0

x1=1

n=1

net_error=1

write(*,*)"iteration |x1 |x2 |f(x2) |net_error"

do while(net_error>1.0E-6)

f_x0=f(x0)

f_x1=f(x1)

x2=x1-((f_x1)*(x0-x1))/(f_x0-f_x1)

f_x2=f(x2)

net_error=abs(x2-x1)

write(*, '(I8,4F18.10)')n,x1,x2,f_x2,net_error

x0=x1

x1=x2

n=n+1

end do

Write(*,*)"the Approximation of root is x2=",x2

contains

Real function f(x)

Real, Intent(In)::x

f=4*x**3-1-exp(x**2/2)

end function

End Program